

Technical note: Fisher-matrix forecasts from ASTRA environment-split clustering

ASTRA clustering project

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Abstract

We forecast cosmological constraints on $\{\omega_b, \omega_{cdm}, n_s, \sigma_8\}$ from the redshift-space two-point clustering of an HOD galaxy sample split by cosmic-web environment with the ASTRA classification. Cosmology derivatives come from the AbacusSummit linear derivative grid; covariances come from $500 h^{-1} \text{Mpc}$ subboxes of the fiducial box. The note documents the full chain of methodological refinements that the project went through, each driven by a concrete failure of the previous step: (i) paired central differences and the discovery that the available HOD catalogs are *not* matched across cosmologies, so the cosmology derivatives are HOD-contaminated; (ii) phase-matched full-box differences (Tier 0) that remove the derivative *noise* but not the contamination *bias*; (iii) an explicit HOD-contamination subtraction calibrated on extra fiducial HOD draws (Tier 1); (iv) a joint cosmology+HOD Fisher that marginalises the HOD nuisances under a prior; (v) a covariance pooled across cosmologies that lifts the Hartlap limit and lets the quadrupole and random-quantile statistics into the data vector; and finally (vi) a global linear response model fit across ~ 50 HOD draws *per* cosmology, whose cosmology derivatives are HOD-clean by construction and which supersedes the contamination-subtraction machinery. The headline methodological results are that the ASTRA environment split, the quadrupole, and the random-quantile statistics each add information beyond the full-sample monopole even after HOD marginalisation.

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1 Fisher formalism

For a Gaussian-distributed data vector ξ with a parameter-independent covariance C , the Fisher information matrix for parameters θ_α is

$$F_{\alpha\beta} = \frac{\partial \xi^\top}{\partial \theta_\alpha} C^{-1} \frac{\partial \xi}{\partial \theta_\beta}, \quad \sigma(\theta_\alpha) = [(F^{-1})_{\alpha\alpha}]^{1/2} \quad (1)$$

[1]. The marginalised error on θ_α uses the inverse of the *full* matrix; the conditional (all other parameters fixed) error is $F_{\alpha\alpha}^{-1/2}$.

The curves shown later are not measured posteriors but the likelihood implied by Eq. (1): the log-likelihood is expanded to second order about the fiducial point, so it is by construction Gaussian,

$$\mathcal{L}(\theta) \propto \exp\left[-\frac{1}{2}(\theta - \theta_{\text{fid}})^\top F(\theta - \theta_{\text{fid}})\right]. \quad (2)$$

For a parameter handled in the logarithm (e.g. $\ln \omega_b$) the physical width follows from the same expansion, $\sigma(\omega_b) = \omega_{b,\text{fid}} \sigma(\ln \omega_b)$.

Our data vectors are built from the redshift-space 2PCF multipoles ($\ell = 0, 2$; 15 bins over $0 < s < 150 h^{-1} \text{Mpc}$) of: the full LRG-like sample; the four ASTRA environment quantiles of the data (Q1 most underdense, Q4 most overdense); the four quantiles of the *random* catalog, which ASTRA classifies on the same Delaunay tessellation and which therefore trace the same cosmic web; and the cross-correlations of the full sample with each quantile. The measurement pipeline (Landy–Szalay estimator) is described in the companion zero-crossing note.

The inverse covariance built from N samples and p bins is biased; we apply the Hartlap correction $(N - p - 2)/(N - 1)$ [3] to C^{-1} throughout. This factor is what makes the number of covariance samples the binding constraint on how long a data vector can be, a theme that recurs below.

2 Simulations and the HOD-mismatch problem

Table 1 lists the nine simulations: the fiducial cosmology and the four $\pm$ pairs of the AbacusSummit linear derivative grid [2], all at phase **ph000**, $z = 0.5$, each populated with an LRG-like HOD catalog ($\sim 4 \times 10^6$ galaxies) from the EMC emulator set. The grid holds the CMB acoustic scale θ_* fixed (h retuned) and, except for the σ_8 pair, retunes A_s so σ_8 is held constant; the cosmology signal in each pair is therefore the response to a *single* parameter.

Cosmology	Variation	Fisher HOD pick	Ensemble
c000	fiducial (Planck 2018 Λ CDM)	hod484	50 draws
c100 / c101	$\pm 2\%$ in $\ln \omega_b$	hod179 / hod152	50 / 50
c102 / c103	$\pm 3.3\%$ in $\ln \omega_{cdm}$	hod556 / hod861	50 / 50
c104 / c105	± 0.01 in n_s	hod498 / hod589	50 / 50
c112 / c113	$\pm 2\%$ in σ_8	hod507 / hod483	50 / 50

Table 1: Simulations of the 4-parameter Λ CDM Fisher design. All are AbacusSummit base boxes ($2000 h^{-1} \text{Mpc}$, phase **ph000**). “Fisher HOD pick” is the single catalog used for the early ($\pm$ -pair) derivatives; “Ensemble” is the per-cosmology set of ~ 50 maximin-selected HOD draws used by the global response model of Section 7.

The core problem. The emulator catalogs are *not* HOD-matched across cosmologies: each cosmology carries its own random draws from the 12-parameter yuan23 HOD prior, with different parameter values at the same `hodNNN` index (verified from all catalog headers; zero exact matches). A derivative formed from a single catalog per cosmology therefore mixes the cosmology signal with the difference in galaxy–halo connection between the two catalogs. Measured directly, the `c100–c101` difference is a scale-dependent -6% to $+9\%$ tilt of ξ_0 , against a $\sim 1\%$ physical ω_b signal at fixed σ_8 and θ_* ; the σ_8 pair difference loses $\sim 89\%$ of its amplitude to a cancelling HOD mismatch. Removing this contamination is the thread that connects every method below.

3 Derivatives

3.1 Paired subbox central differences

The first derivative estimator differences the *same* $500 h^{-1}$ Mpc subbox i in the two cosmologies of a pair before averaging,

$$\frac{\partial \xi_i}{\partial \theta} = \frac{\xi_i(\theta + \Delta\theta) - \xi_i(\theta - \Delta\theta)}{2 \Delta\theta}. \quad (3)$$

Because the grid simulations share the fiducial phases and our pipelines share random seeds, cosmic variance largely cancels in Eq. (3); the mean over the 64 subboxes gives the derivative and its scatter the uncertainty. Figure 1 shows these derivatives for all four parameters and all measured statistics.

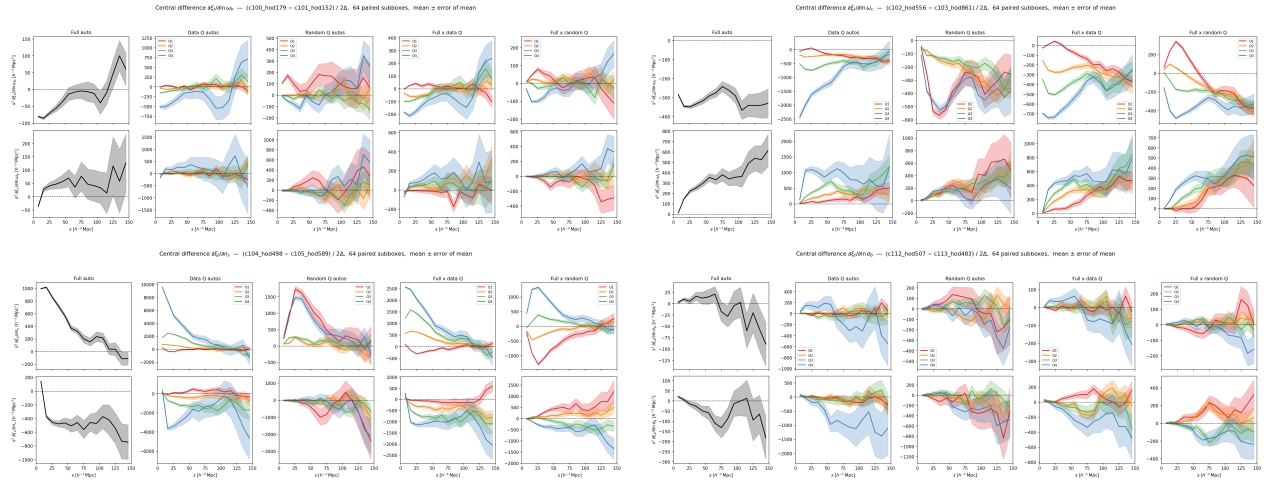


Figure 1: Paired-subbox central-difference derivatives $\partial \xi_\ell / \partial \theta$ for $\theta = \ln \omega_b$ (top left), $\ln \omega_{\text{cdm}}$ (top right), n_s (bottom left), $\ln \sigma_8$ (bottom right). Columns within each panel: full auto, data quantile autos, random quantile autos, full $\times$ data quantiles, full $\times$ random quantiles; rows: monopole and quadrupole. Lines are the mean over 64 paired subboxes, bands the error of the mean.

3.2 Tier 0: phase-matched full-box differences

A subbox has open boundaries and high shot noise. Differencing the *whole* $2000 h^{-1}$ Mpc box instead,

$$\frac{\partial \xi}{\partial \theta} = \frac{\xi(\theta + \Delta\theta) - \xi(\theta - \Delta\theta)}{2 \Delta\theta}, \quad \text{Var} \left(\frac{\partial \xi}{\partial \theta} \right)_b = \frac{\text{std}_+^2 / n_+ + \text{std}_-^2 / n_-}{(2 \Delta\theta)^2}, \quad (4)$$

where the noise model uses the scatter over the $n = 3$ ASTRA-random iterations of each fixed box (the only part that does not cancel in the phase-matched difference). Figure 2 compares the subbox

and full-box derivatives through the resulting single-parameter σ , evaluated against a common covariance. For the three already-clean parameters the two methods agree to $\approx 3\%$ (consistency check); for σ_8 the full-box derivative removes the derivative *noise* that had dominated the subbox estimate, tightening $\sigma(\ln \sigma_8)$ by $\sim 1.8\times$ and reducing its noise share from 79% to $\approx 0\%$. Tier 0 fixes derivative noise but *not* the HOD contamination bias of Section 2.

3.3 Tier 1: explicit HOD-contamination subtraction

To remove the contamination *bias*, the HOD response $\partial\xi/\partial\theta_{\text{HOD}}$ is measured by regressing the full-box ξ on the 12 standardised HOD parameters across ~ 50 extra fiducial-cosmology HOD draws, and the contamination of each pair is subtracted,

$$\left.\frac{\partial\xi}{\partial\theta}\right|_{\text{corr}} = \left.\frac{\partial\xi}{\partial\theta}\right|_{\text{raw}} - \frac{\Delta\theta_{\text{HOD}} \cdot \mathbf{G}}{2\Delta\theta}, \quad \mathbf{G} = \frac{\partial\xi}{\partial\theta_{\text{HOD}}}, \quad (5)$$

where $\Delta\theta_{\text{HOD}}$ is the measured HOD-parameter mismatch between the two catalogs of the pair. Figure 3 shows the raw, contamination, and corrected full-auto derivatives. The contamination is a large, smooth, sign-changing tilt comparable to or larger than the raw derivative, confirming the derivatives were heavily contaminated. As a physical check, linear theory gives $\partial\xi/\partial\ln\sigma_8 = 2\xi$ for the full sample; the raw full-box σ_8 derivative is only 11% of 2ξ (the HOD mismatch had cancelled most of the signal), while the corrected derivative recovers 136% of 2ξ — back in the right regime.

4 Covariance

The covariance enters Eq. (1) as the normalised correlation matrix $R_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$ shown in Fig. 4. Neighbouring separation bins are strongly correlated, with the correlation length growing toward large s . The full auto and the full $\times$ data Q4 blocks are nearly perfectly correlated (the redundancy discussed below), whereas the full $\times$ random Q1 block is *anti*-correlated with them — the only genuinely complementary information among those three.

Pooling across cosmologies. Estimating $\mathbf{C}$ from the 64 fiducial subboxes alone caps the data vector at $p < 64$ bins; adding the quadrupole to the quantile vectors exceeds this and drives the Hartlap factor negative. Since the covariance varies far more slowly with cosmology than the mean does, we instead pool the mean-subtracted subboxes across all nine cosmologies,

$$\mathbf{C}_{\text{sub}} = \frac{1}{N - n_c} \sum_{c=1}^{n_c} \sum_{i=1}^{64} (\xi_{ci} - \bar{\xi}_c)(\xi_{ci} - \bar{\xi}_c)^\top, \quad N = n_c \times 64 = 576, \quad (6)$$

removing each cosmology’s own mean $\bar{\xi}_c$ so that only cosmic-variance fluctuations survive. The full-box covariance is $\mathbf{C} = \mathbf{C}_{\text{sub}}/64$. With $N = 576$ the Hartlap factor stays $\gtrsim 0.75$ even at $p = 144$ bins, so mono+quad data and random quantile vectors fit comfortably. The caveat is that subboxes tile a single box per cosmology and share large-scale modes, so the effective number of independent samples is below 576 — but this approximation was already implicit in the 64-subbox estimate.

5 Joint cosmology + HOD Fisher

Fixing the HOD exactly is optimistic, because the σ_8 response is amplitude-like and so degenerate with HOD bias. We therefore build one Fisher over the 4 cosmology parameters *and* the 12 HOD

parameters and marginalise the latter,

$$F = D C^{-1} D^T + F_{\text{prior}}, \quad D = \begin{bmatrix} \partial \xi / \partial \theta_{\text{cosmo}} \text{ (4 rows)} \\ \partial \xi / \partial \theta_{\text{HOD}} \text{ (12 rows)} \end{bmatrix}, \quad F_{\text{prior}} = \text{diag}(\mathbf{0}_4, \sigma_{\text{prior}}^{-2}), \quad (7)$$

where the cosmology rows are the HOD-corrected derivatives (Eq. 5) and the HOD rows are the regression gradient $\mathbf{G}$. The poorly-constrained HOD directions are regularised by a Gaussian yuan23 prior (σ_{prior} = spread of the fiducial HOD draws), *not* by truncation, which would be an implicit infinite prior and reintroduce the fixed-HOD optimism. The conditional cosmology error inverts the 4×4 cosmology block of $D C^{-1} D^T$; the marginalised error inverts the full 16×16 F and reads off the cosmology block.

Table 2 gives the full-auto result. n_s and ω_b degrade most under marginalisation because their tight conditional errors overlap the broadband HOD tilt directions; in fractional terms all four parameters land at 0.6–1.2% from a single $2 h^{-1} \text{Mpc}[\text{Gpc}]$ box. Figure 5 shows the degradation, Fig. 5 the robustness check (the cosmology errors plateau by ~ 4 marginalised HOD directions, so the prior-controlled tail is irrelevant), and Fig. 6 the conditional vs. marginalised ellipses.

Parameter	σ_{cond} (HOD fixed)	σ_{marg} (HOD marg.)	degrade	$\sigma_{\text{marg}}/\theta_{\text{fid}}$
ω_b	3.08×10^{-5}	2.49×10^{-4}	$8.1 \times$	1.1%
ω_{cdm}	4.88×10^{-4}	1.01×10^{-3}	$2.1 \times$	0.85%
n_s	5.78×10^{-4}	5.89×10^{-3}	$10.2 \times$	0.61%
$\ln \sigma_8$	3.18×10^{-3}	9.31×10^{-3}	$2.9 \times$	1.15%

Table 2: Joint Fisher on the full-auto mono+quad vector (HOD-corrected derivatives, 64-subbox covariance scaled to the full box). Conditional holds the HOD fixed; marginalised marginalises the 12 HOD nuisances under the yuan23 prior.

6 Which data vector? Quadrupole and random quantiles

With the pooled covariance (Eq. 6) lifting the Hartlap limit, we can ask which statistics carry independent information *after* HOD marginalisation. Table 3 compares vectors that share the same monopole binning and simply add the quadrupole.

Data vector	p	Hartlap	$\sigma(\omega_b)$	$\sigma(\omega_{\text{cdm}})$	$\sigma(n_s)$	$\sigma(\sigma_8)$
full auto (mono+quad)	30	0.95	2.18	9.31	52.3	80.4
data Q autos (mono)	32	0.94	1.94	7.53	29.9	86.1
data Q autos (mono+quad)	64	0.89	1.46	5.51	18.1	54.5
full+data Q autos (mono)	40	0.93	1.68	5.80	24.0	68.3
full+data Q autos (mono+quad)	80	0.86	1.14	4.47	15.5	46.0

Table 3: HOD-marginalised σ (units: $\omega_b, \omega_{\text{cdm}}$ in 10^{-4} ; n_s in 10^{-4} ; σ_8 in 10^{-4}), pooled 576-sample covariance, HOD-corrected derivatives. Adding the quadrupole at fixed monopole binning tightens every parameter by 1.3–1.65 $\times$; the earlier finding that “the quadrupole adds almost nothing” was a Hartlap-budget artifact of the 64-subbox covariance.

The full set of marginalised constraints by data vector is shown in the corner plot of Fig. 7. Two robust conclusions: the quadrupole carries real, largely independent information once it is not

crowded out of the Hartlap budget; and the ASTRA environment split itself helps — the quantile autos beat the full auto, and combining the full sample with the quantiles is tighter still.

The single-parameter view of the same comparison is shown in Fig. 8, where each candidate data vector is drawn as a one-parameter Fisher Gaussian (Eq. 2) around the fiducial value, for one subbox volume and for the full box. These reproduce the per-vector ranking parameter by parameter and make the relative widths directly legible.

7 Global response model

The contamination subtraction of Section 3 is a first-order patch. The cleaner route, enabled by the ~ 500 existing HOD draws *per* cosmology, is to run ~ 50 maximin-selected draws in every cosmology and fit one global linear response across all runs,

$$\boldsymbol{\xi}(\boldsymbol{\theta}) \approx \boldsymbol{\xi}_0 + \sum_p a_p \theta_{\text{cosmo},p} + \sum_q b_q \theta_{\text{HOD},q}. \quad (8)$$

Because the HOD term absorbs the HOD variation explicitly, the cosmology coefficients a_p are the HOD-marginalised cosmology derivatives: the cross-cosmology HOD mismatch no longer contaminates them, and (all runs sharing **ph000**) cosmic variance cancels in a_p . The HOD coefficients b_q give the gradient $\mathbf{G}$. This single fit replaces the Tier 0+Tier 1 chain; the finite-difference derivatives are retained only as a cross-check (Fig. 9).

The joint Fisher of Eq. (7) then consumes the global derivatives, and we report two marginalisation routes that should agree: route (a), the 16-parameter joint Fisher with the yuan23 prior; and route (b), a prior-free check that adds an empirical HOD covariance $\mathbf{C}_{\text{HOD}}$ — the sample covariance of the data vector across the same-phase fiducial HOD ensemble (pure HOD scatter, cosmic variance cancelled) — to the cosmic-variance covariance, $\mathbf{C}_{\text{tot}} = \mathbf{C}_{\text{CV}} + \mathbf{C}_{\text{HOD}}$, and uses only the 4 cosmology rows. Table 4 gives the current results, spanning data and random quantiles, monopole and quadrupole.

Data vector	p	Hartlap	$\sigma(\omega_b)$	$\sigma(\omega_{cdm})$	$\sigma(n_s)$	$\sigma(\sigma_8)$
full auto (mono+quad)	30	0.95	2.12	9.74	52.6	81.5
data Q autos (mono)	32	0.94	2.02	7.09	29.5	83.6
data Q autos (mono+quad)	64	0.89	1.45	5.59	17.7	53.9
rand Q autos (mono+quad)	64	0.89	1.35	5.81	21.4	63.4
data+rand Q autos (mono+quad)	128	0.78	0.86	3.09	11.2	40.8
full+data+rand Q autos (mono+quad)	144	0.75	0.69	2.72	9.76	36.5

Table 4: Global-response (route a) HOD-marginalised σ (units: $\omega_b, \omega_{cdm}, n_s$ in 10^{-4} ; σ_8 in 10^{-4}), pooled 576-sample covariance. The two marginalisation routes agree to $1.3\text{--}1.6\times$ (route b is looser, being prior-free with only 50 $\mathbf{C}_{\text{HOD}}$ draws). These numbers are a smoke test on the 58 runs presently on disk (50 fiducial + 1 per off-fiducial cosmology); the design-matrix condition number is 1.9 and $\partial\xi/\partial\ln\sigma_8$ is $0.68 \times 2\xi$, both of which improve as the ~ 50 -per-cosmology campaign completes.

Adding the random quantiles and the quadrupole both help: the random quantile autos are only slightly weaker than the data quantile autos alone, but combining data, random, and the full sample with the quadrupole is the tightest vector on every parameter, because the random quantiles respond to cosmology and the HOD differently from the galaxies and so break degeneracies the galaxy clustering alone cannot.

8 Status and next steps

The pipeline is complete and the global response model (Eq. 8) is the current method. The ~ 50 -draw-per-cosmology campaign (~ 392 full-box runs) is in progress; when it finishes, the global derivatives tighten and the $\partial\xi/\partial\ln\sigma_8 = 2\xi$ sanity ratio should approach unity. Remaining limitations are: the covariance still rests on partially-correlated subboxes (only one phase exists); the global fit is linear over the HOD prior (a quadratic/interaction extension would test cosmology-dependence of the HOD response); and derivative-noise and gradient-uncertainty are not yet propagated into the Fisher.

References

- [1] M. Tegmark, A. N. Taylor, A. F. Heavens, *Karhunen-Loève eigenvalue problems in cosmology: how should we tackle large data sets?*, ApJ 480, 22 (1997).
- [2] N. A. Maksimova et al., *AbacusSummit: a massive set of high-accuracy, high-resolution N-body simulations*, MNRAS 508, 4017 (2021).
- [3] J. Hartlap, P. Simon, P. Schneider, *Why your model parameter confidences may be too optimistic. Unbiased estimation of the inverse covariance matrix*, A&A 464, 399 (2007).

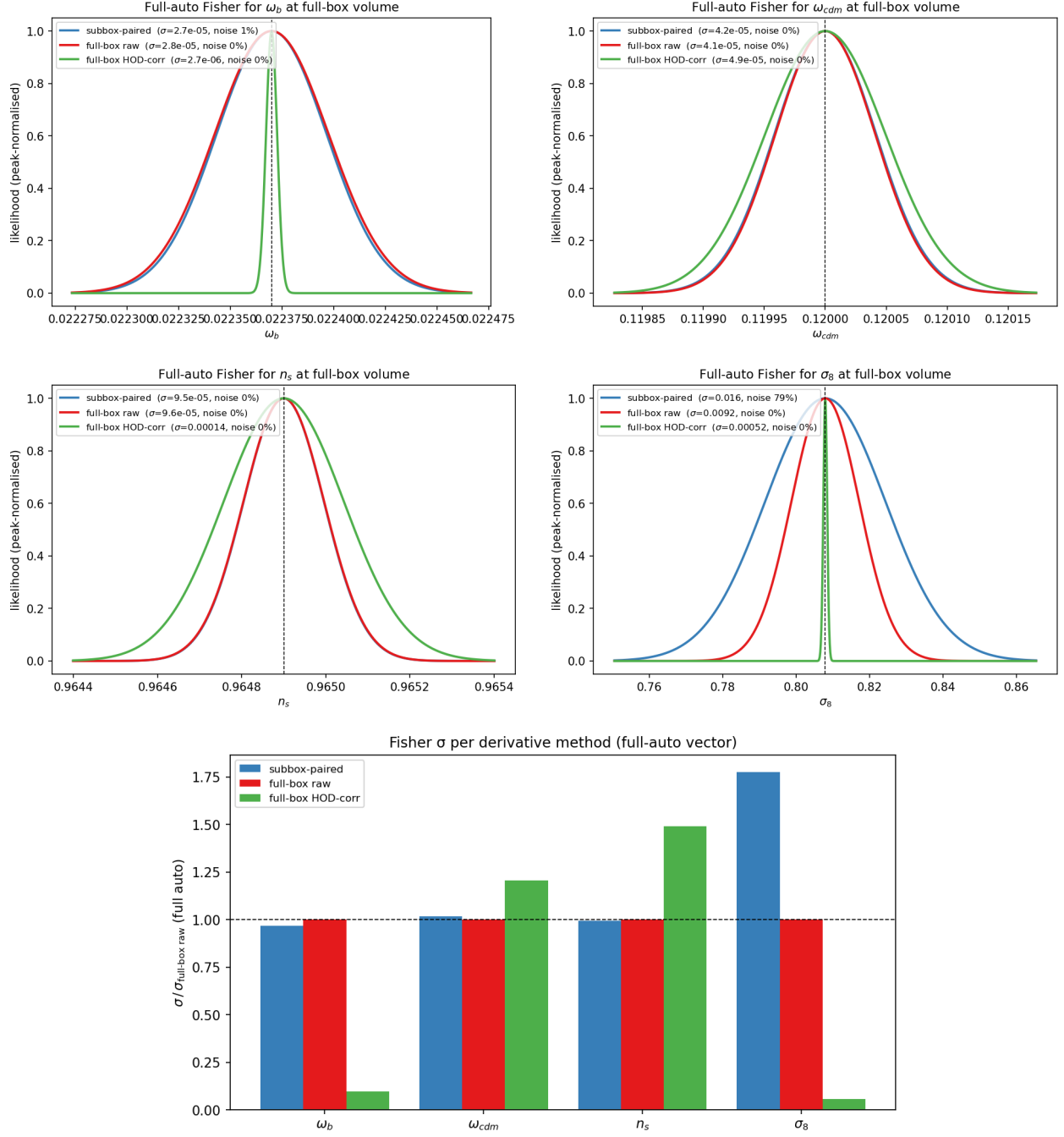


Figure 2: Tier 0: subbox vs. full-box derivative for each parameter (and summary, bottom). The two derivative sources are evaluated against the same covariance, so only the numerator and noise differ. The σ_8 panel shows the noise-driven gain.

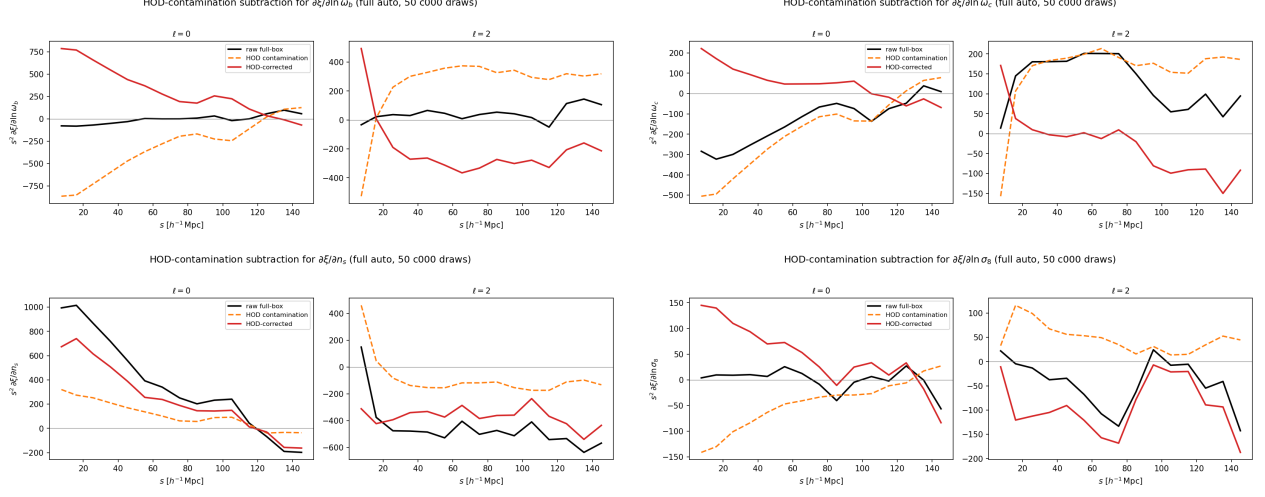


Figure 3: Tier 1 HOD-contamination subtraction, Eq. (5), for the full-sample auto-correlation ($\ell = 0$ and $\ell = 2$). Black: raw full-box derivative; orange dashed: modelled HOD contamination; red: corrected derivative.

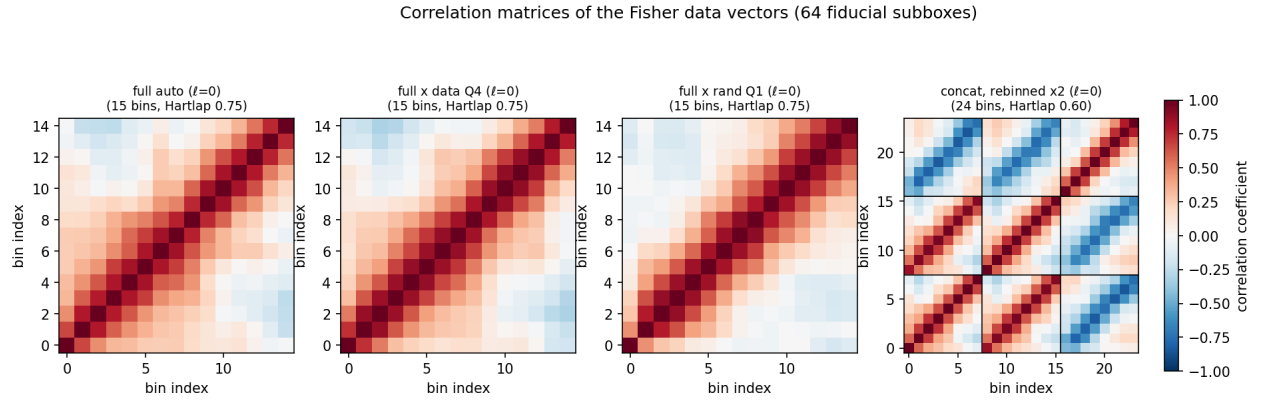


Figure 4: Correlation matrices of representative data vectors, estimated from the fiducial subboxes. Panel titles give the number of bins and the Hartlap factor $(N - p - 2)/(N - 1)$.

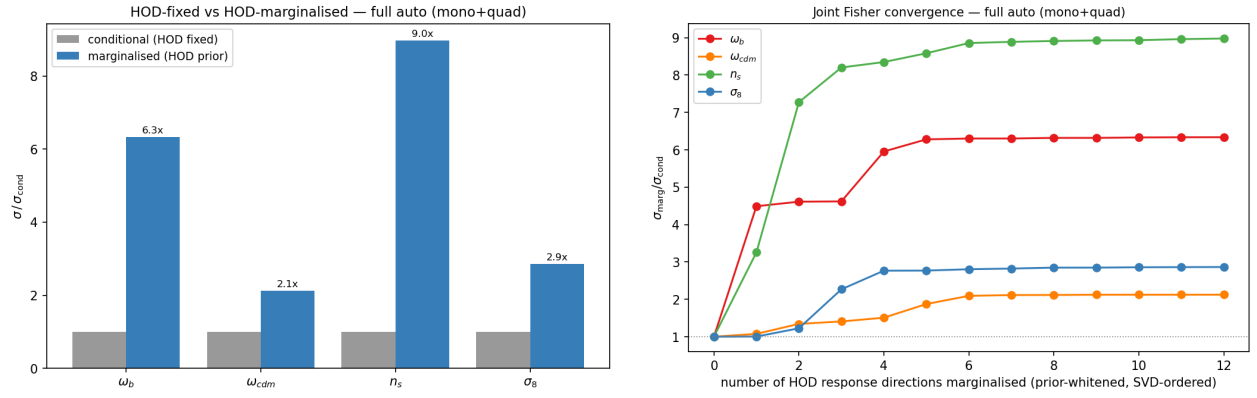


Figure 5: Left: cosmology errors, HOD-fixed vs. HOD-marginalised (the degradation factors of Table 2). Right: convergence of the marginalised error as HOD response directions are marginalised one at a time (prior-whitened, SVD-ordered); the plateau by $k \approx 4$ shows the HOD response is effectively low-rank.

Joint Fisher (68% solid / 95% dashed): HOD-fixed (grey) vs HOD-marginalised (blue) — full auto (mono+quad)

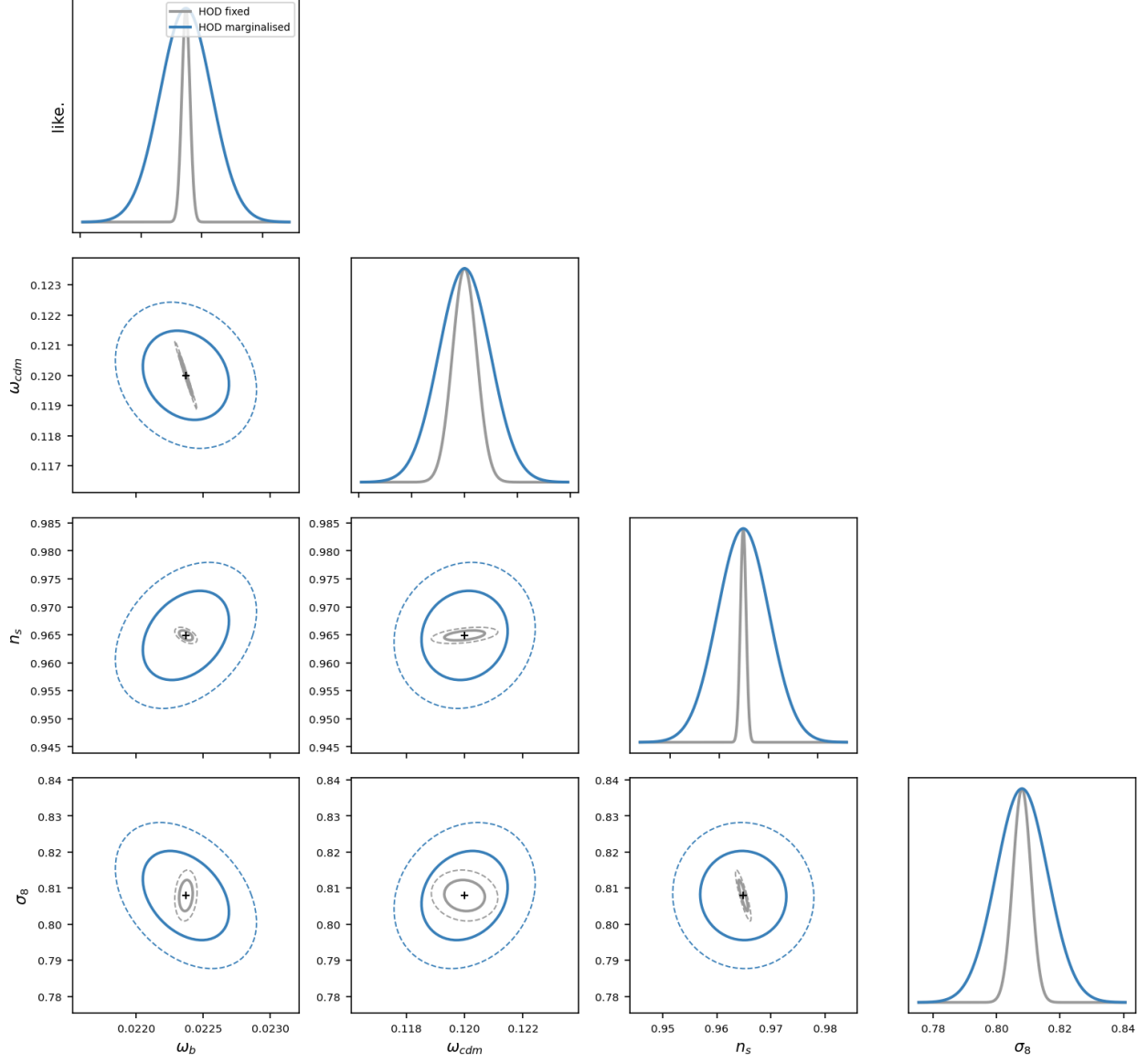


Figure 6: Joint Fisher 68%/95% ellipses in physical units: HOD-fixed (grey) vs. HOD-marginalised (blue). The blue ellipses dwarf the grey and tilt to show the cosmology–cosmology degeneracies that HOD marginalisation opens up.

HOD-marginalised cosmology constraints by data vector (68% solid / 95% dashed)

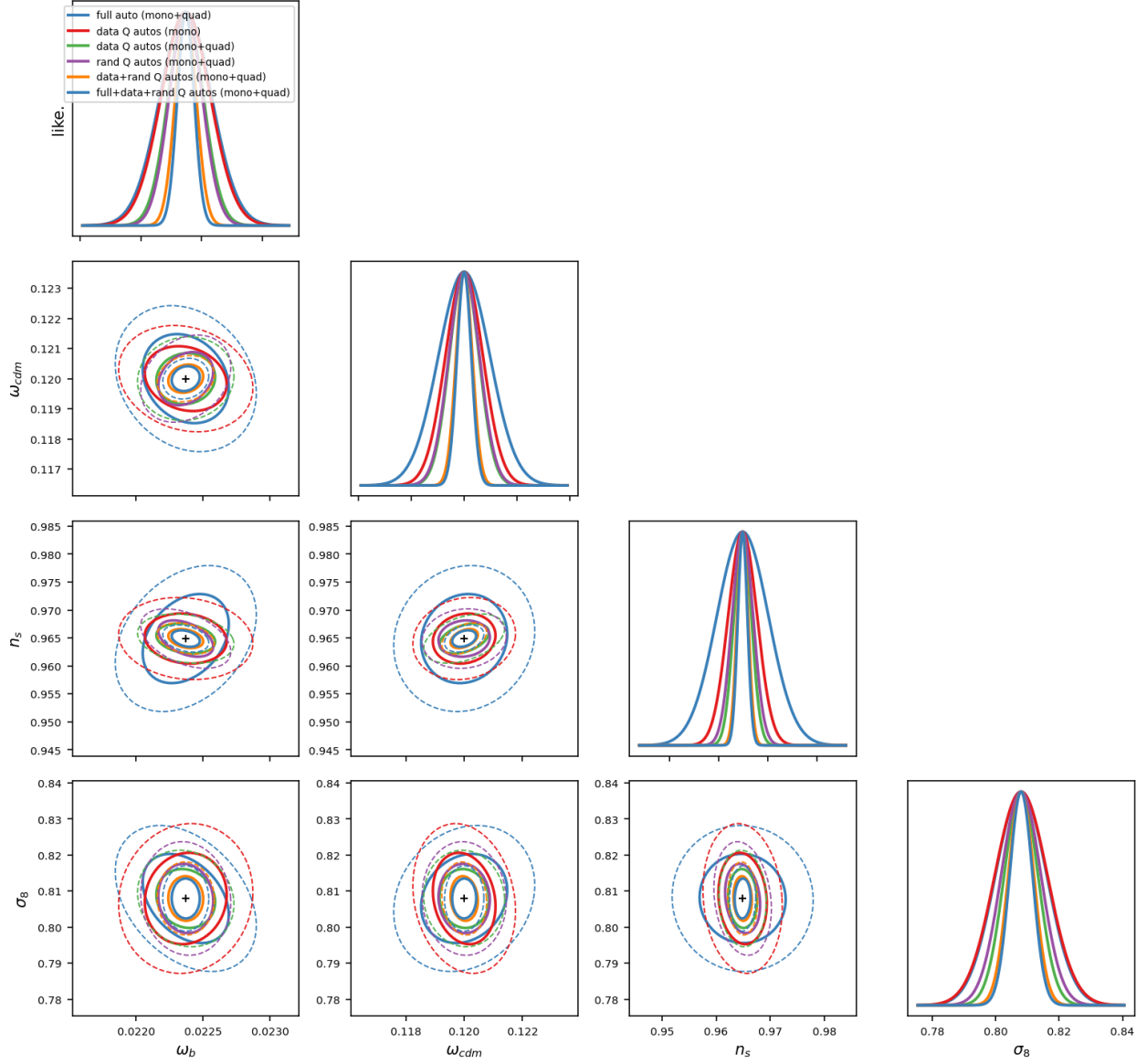
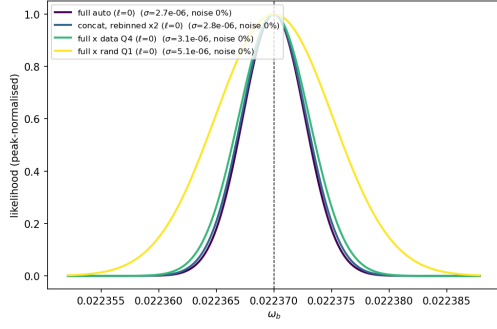
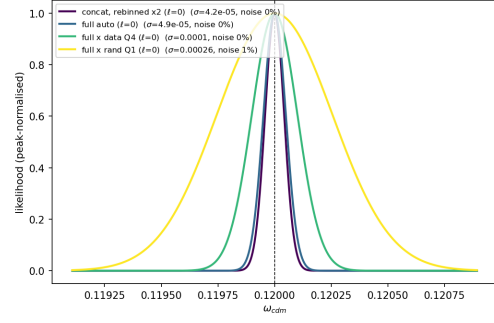


Figure 7: HOD-marginalised 68%/95% cosmology constraints by data vector. Vectors that add the ASTRA quantiles and the quadrupole shrink the ellipses relative to the full-auto baseline.

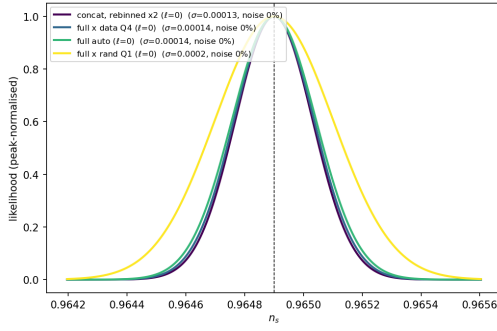
Fisher forecasts for ω_b per data vector (HOD-corrected full-box derivative, full 2000 Mpc/h volume, Hartlap-corrected)



Fisher forecasts for $\ln \omega_{cdm}$ per data vector (HOD-corrected full-box derivative, full 2000 Mpc/h volume, Hartlap-corrected)



Fisher forecasts for n_s per data vector (HOD-corrected full-box derivative, full 2000 Mpc/h volume, Hartlap-corrected)



Fisher forecasts for σ_8 per data vector (HOD-corrected full-box derivative, full 2000 Mpc/h volume, Hartlap-corrected)

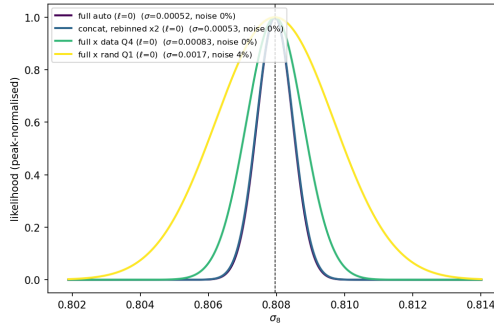


Figure 8: Single-parameter Fisher likelihoods (Eq. 2), normalised to unit peak, for a selection of data vectors and each of the four parameters: $\ln \omega_b$ (top left), $\ln \omega_{cdm}$ (top right), n_s (bottom left), $\ln \sigma_8$ (bottom right). Narrower is more constraining.

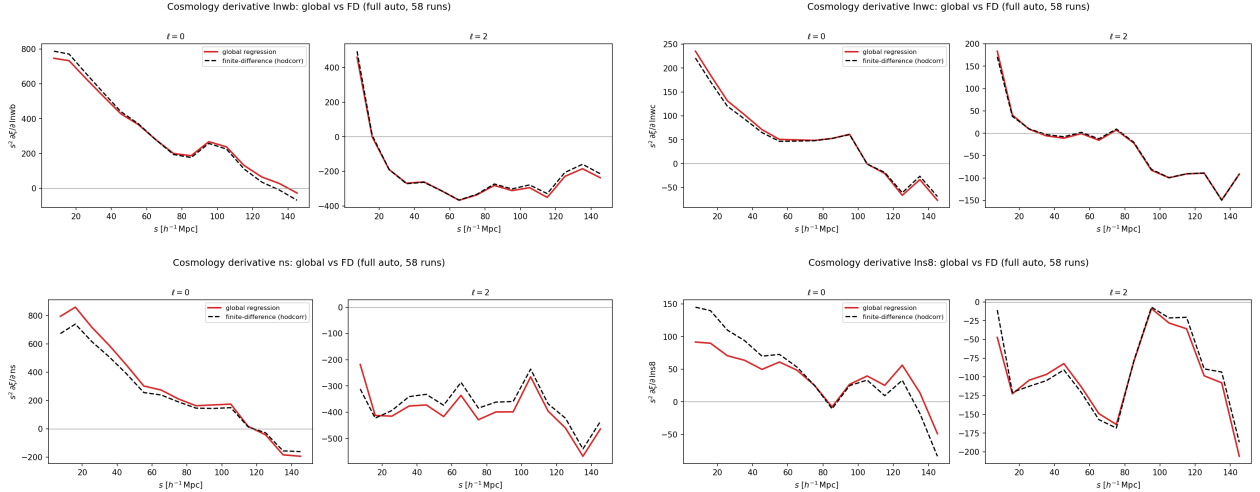


Figure 9: Cross-check of the global-regression cosmology derivative (Eq. 8, red) against the finite-difference HOD-corrected derivative (Eq. 5, black dashed) for the full-sample auto, $\ell = 0$ and $\ell = 2$.